

School of Advanced Computing & Information Technology

Department of Computer Science, Gujarat University

SEC-316 Advanced Python Programming (Practical)

Assignment - Unit 1

- **Pandas**

1. What does Pandas stand for?
2. Name any two Python libraries on which Pandas is built.
3. By default, what is the index of a Pandas Series?
4. How do you check if a Series is empty?
5. Give one difference between Series and DataFrame.
6. Which method is used to display the first 5 rows of a DataFrame?
7. What is the default axis for operations like *sum()* in Pandas?
8. Name two methods to handle missing values in Pandas.
9. Create a Pandas Series from a dictionary with countries as keys and capitals as values.
10. Write code to access the 3rd element from a Series using both positional and labelled indexing.
11. Differentiate between *.head()* and *.tail()* functions in Pandas with an example.
12. How can you change the index of a Pandas Series? Write an example.
13. Explain the use of *add()* function with *fill_value* parameter.
14. Write a Python statement to count the number of non-null values in a Series.
15. How can we create a DataFrame from a list of dictionaries? Give a small example.
16. What will happen if the length of index and values mismatch while creating a Series?
17. Write the output of the following code:

```
import pandas as pd
s=pd.Series([12, None, 10])
print(s.count())
```

18. Give two differences between *loc[]* and *iloc[]* in Pandas.
19. What do you mean by aggregation in Pandas? Give two functions used for aggregation.
20. Write a program to demonstrate indexing and slicing in Series using both labels and positions.

21. Perform the following mathematical operations on two Series: Addition, Subtraction, Multiplication, Division (with missing values handling).
22. Write code to calculate the mean, median, mode, variance, and standard deviation of marks in a DataFrame.
23. Explain the use of quantile() function with examples of 25%, 50%, and 75% quartiles.
24. Write a program to demonstrate handling missing values in a DataFrame using fillna() and dropna().
25. Explain describe() function in Pandas with proper output.
26. Write code to export a DataFrame to CSV and then import it back.
27. Write a case study (example data) where you use Group By to find the average marks of students per subject.

• Matplotlib

1. What is Matplotlib in Python?
2. Which function is used to plot a simple line graph?
3. Write the code to display a graph window in Matplotlib.
4. What does `plt.xlabel()` do?
5. Differentiate between `plt.plot()` and `plt.scatter()`.
6. Write code to plot a line graph for `x = [1,2,3,4]` and `y = [2,4,6,8]`.
7. Plot a bar chart showing marks of 4 students: `["A", "B", "C", "D"] = [80, 65, 90, 70]`.
8. Plot a histogram for the dataset `[5,10,15,10,5,20,25,20,15,10]`.
9. Draw a scatter plot for `x=[5,7,8,7,6,9,5,7]` & `y=[99,86,87,88,100,86,103,87]`
10. Write code to draw a pie chart showing the share of fruits: Apple = 40, Banana = 25, Orange = 20, Grapes = 15.
11. Write code to draw both less-than and greater-than ogives on the same plot for comparison.
12. Using Matplotlib, plot a ogive for the dataset: `Data = [7,8,5,6,9,7,5,8,6,7,9,8,6,5,7,8,9,7,6,8]`

13. Make Bar Graph and Pie chart for the following data.

TABLE 2.1 DATA FROM A SAMPLE OF 50 SOFT DRINK PURCHASES

Coke Classic	Sprite	Pepsi
Diet Coke	Coke Classic	Coke Classic
Pepsi	Diet Coke	Coke Classic
Diet Coke	Coke Classic	Coke Classic
Coke Classic	Diet Coke	Pepsi
Coke Classic	Coke Classic	Dr. Pepper
Dr. Pepper	Sprite	Coke Classic
Diet Coke	Pepsi	Diet Coke
Pepsi	Coke Classic	Pepsi
Pepsi	Coke Classic	Pepsi
Coke Classic	Coke Classic	Pepsi
Dr. Pepper	Pepsi	Pepsi
Sprite	Coke Classic	Coke Classic
Coke Classic	Sprite	Dr. Pepper
Diet Coke	Dr. Pepper	Pepsi
Coke Classic	Pepsi	Sprite
Coke Classic	Diet Coke	

DH	CSI	DH	CSI	L&O
Trace	CSI	L&O	Trace	CSI
CSI	DH	Trace	CSI	DH
L&O	L&O	L&O	CSI	DH
CSI	DH	DH	L&O	CSI
DH	Trace	CSI	Trace	DH
DH	CSI	CSI	L&O	CSI
L&O	CSI	Trace	Trace	DH
L&O	CSI	CSI	CSI	DH
CSI	DH	Trace	Trace	L&O

Brown	Williams	Williams	Williams	Brown
Smith	Jones	Smith	Johnson	Smith
Davis	Smith	Brown	Williams	Johnson
Johnson	Smith	Smith	Johnson	Brown
Williams	Davis	Johnson	Williams	Johnson
Williams	Johnson	Jones	Smith	Brown
Johnson	Smith	Smith	Brown	Jones
Jones	Jones	Smith	Smith	Davis
Davis	Jones	Williams	Davis	Smith
Jones	Johnson	Brown	Johnson	Davis

14. Make Histogram, Line chart, Area Chart, Scatter Diagram and Ogive for the following data.

Consider the following data.

14	21	23	21	16
19	22	25	16	16
24	24	25	19	16
19	18	19	21	12
16	17	18	23	25
20	23	16	20	19
24	26	15	22	24
20	22	24	22	20

- Develop a frequency distribution using classes of 12–14, 15–17, 18–20, 21–23, and 24–26.
- Develop a relative frequency distribution and a percent frequency distribution using the classes in part (a).

187	184	174	185
175	172	202	197
165	208	215	164
162	172	182	156
172	175	170	183

1200	850	740	590	340
450	890	260	610	350
1780	180	850	2050	770
800	1090	510	520	220
1450	280	1120	200	350

4.1	1.5	10.4	5.9	3.4	5.7	1.6	6.1	3.0	3.7
3.1	4.8	2.0	14.8	5.4	4.2	3.9	4.1	11.1	3.5
4.1	4.1	8.8	5.6	4.3	3.3	7.1	10.3	6.2	7.6
10.8	2.8	9.5	12.9	12.1	0.7	4.0	9.2	4.4	5.7
7.2	6.1	5.7	5.9	4.7	3.9	3.7	3.1	6.1	3.1

- **Network Programming**

1. Define a network and mention two advantages of networking.
2. What is the difference between TCP and UDP?
3. What is the role of an IP address in a network? Give an example.
4. Explain what a port number is.
5. Name two Python modules used for network programming.
6. Write the syntax for creating a TCP socket in Python.
7. What does the function `socket.gethostbyname('www.google.com')` return?
8. Differentiate between `encode()` and `decode()` in socket communication.
9. Which Python module is used to download web pages or images from the Internet?
10. Write one line of Python code to parse the URL `https://www.python.org/`.
11. Explain the layers of the TCP/IP model with an example of how a web page is transmitted.
12. Write a Python program to find the IP address of `www.gmail.com`.
13. With an example, explain how `urllib.parse.urlparse()` can extract different parts of a URL.
14. Write a Python program to download the source code of `https://www.python.org/`.

15. Explain the steps involved in creating a TCP server program in Python.
16. Write a Python program to download and save the homepage of <https://www.python.org/> into your system.
17. Write a Python program to download any image from the Internet using `urllib.request.urlretrieve()`.
18. Develop a TCP server program that sends a welcome message to the client. Write the corresponding client program that receives and prints it.
19. Create a Python file server that sends the contents of a text file to a client when the client provides the file name. Write both server and client programs.
20. Develop a simple chat application in Python where the server and client can exchange messages until the user types "exit".